

		20 N
11	D	At time $\frac{t}{2}$, vertical distance fallen by object, $s_y = u_y t + \frac{1}{2} a_y t^2$ $= \frac{1}{2} g \left(\frac{t}{2} \right)^2$ $= \frac{1}{4} \left(\frac{1}{2} g t^2 \right)$ which means object has fallen a distance of $\frac{1}{4} h$. Hence, P.E. of object is $\frac{3}{4} E$, K.E. of object is $\frac{1}{4} E$.
12	A	If the subtended angle θ is given in degrees, the formula for arc length